The Binomial Simplification Reference

Organized by shape you'd write in brute force → what it collapses to. No proofs. Use it by pattern-matching the shape of your sum.

A note on notation: $n^{\underline{m}} = n(n-1)\cdots(n-m+1)$ is the falling factorial. $[P]$ is 1 if P holds, else 0. Unless stated, k ranges over all integers where terms are nonzero.

0. The Rewriting Moves

These aren't "identities to look up" they're the edits you apply to reshape a sum until it matches something in the tables below. Learn these cold; everything else is lookup.

Move	Rule	Use it when
Symmetry	$\binom{n}{k} = \binom{n}{n-k}$	the summation index is in the wrong slot
Absorption	$k\binom{n}{k} = n\binom{n-1}{k-1}$	there's a loose factor of k kills it
Generalized absorption	$k^{\underline{m}}\binom{n}{k} = n^{\underline{m}}\binom{n-m}{k-m}$	a polynomial in k multiplies a binomial
Downward absorption	$\frac{1}{k+1}\binom{n}{k} = \frac{1}{n+1}\binom{n+1}{k+1}$	k is trapped in the denominator (e.g., expected values)
Trinomial revision	$\binom{n}{m}\binom{m}{k} = \binom{n}{k}\binom{n-k}{m-k}$	"choose a set, then a subset" re-associates the choice
Upper negation	$\binom{-n}{k} = (-1)^k\binom{n+k-1}{k}$	you need to turn an alternating sum into a positive one, or vice versa

Two more that matter:

$$\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1} = \frac{n}{n-k}\binom{n-1}{k} \qquad \binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

The last one is how you turn a formula into an $O(n^2)$ DP when no closed form exists.

Absorption is the workhorse. Any $\sum_k P(k)\binom{n}{k}(\cdots)$ with P a polynomial: expand P in falling factorials $k^{\underline{m}}$ (not in powers k^m), apply generalized absorption to each term, and each becomes a plain shifted binomial sum. This single technique handles most "weighted" variants you'd otherwise look up.

1. The Vandermonde Family Convolution Shapes

The signature: two binomials whose lower indices sum to a constant. This is the most common $O(n^2) \to O(1)$ collapse in CP.

$$\sum_k \binom{m}{k}\binom{n}{p-k} = \binom{m+n}{p} \tag{V}$$

$$\sum_k \binom{m}{k}\binom{n}{p+k} = \binom{m+n}{m+p} \qquad \text{(V-reversed, needs } n \geq 0\text{)}$$

Special case worth memorizing separately (it's V-reversed with $p = 0$):

$$\sum_k \binom{m}{k}\binom{n}{k} = \binom{m+n}{m} \qquad \sum_k \binom{n}{k}^2 = \binom{2n}{n}$$

Weighted variants

$$\sum_k k\binom{m}{k}\binom{n}{p-k} = m\binom{m+n-1}{p-1} \qquad \sum_k k^{\underline{j}}\binom{m}{k}\binom{n}{p-k} = m^{\underline{j}}\binom{m+n-j}{p-j}$$

$$\sum_k k\binom{n}{k}^2 = n\binom{2n-1}{n-1} \qquad \sum_k k^2\binom{n}{k}^2 = n^2\binom{2n-2}{n-1}$$

You don't need to memorize the weighted row apply absorption first, then plain Vandermonde. That's the whole derivation.

Alternating Vandermonde

$$\sum_k (-1)^k \binom{m}{k}\binom{n}{p-k} = (-1)^p \binom{n-m}{p-m} \quad \text{(via upper negation)}$$

In practice: push the $(-1)^k$ into one binomial with upper negation, then apply (V). Don't memorize the result, memorize the route.

Multivariable / multinomial Vandermonde

$$\sum_{k_1+\cdots+k_r=p} \binom{n_1}{k_1}\cdots\binom{n_r}{k_r} = \binom{n_1+\cdots+n_r}{p} \qquad \sum_{i+j+k=n} \binom{n}{i,j,k} x^i y^j z^k = (x+y+z)^n$$

Triple products: apply (V) twice, innermost first. That turns $O(n^3)$ into $O(n)$ or $O(1)$.

2. The "Split a Line" Identity The Hockey-Stick Generalization

This one is underrated and it's the highest-value single formula on this page for CP:

$$\boxed{\sum_{k=0}^{l} \binom{l-k}{m}\binom{q+k}{n} = \binom{l+q+1}{m+n+1}}$$

Recognize it as: "I pick a split point, choose m things left of it and n things right of it." Any sum of that shape is one binomial. It's what collapses most "sum over the position of a pivot" brute forces.

Equivalent common phrasings (including Upper-Index Vandermonde):

$$\sum_{k=0}^{n} \binom{k}{a}\binom{n-k}{b} = \binom{n+1}{a+b+1} \qquad \sum_i \binom{a+i}{i}\binom{b+n-i}{n-i} = \binom{a+b+n+1}{n}$$

Hockey stick (the degenerate cases)

$$\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1} \qquad \sum_{i=0}^{n} \binom{r+i}{i} = \binom{r+n+1}{n}$$

Recognize: a prefix sum down a column of Pascal's triangle collapses. A prefix sum across a row does not (see §5).

3. Alternating Sums The Vanishing Theorem

$$\boxed{\sum_{k=0}^{n} (-1)^k \binom{n}{k} P(k) = 0 \quad \text{whenever } \deg P < n}$$

and if $\deg P = n$ with leading coefficient c, the sum is $(-1)^n c\, n!$.

This is the single most powerful "the answer is zero" tool. Whole $O(n^2)$ sums vanish. If your inner sum is alternating with a polynomial weight, check the degree first before doing any work.

More generally, $\sum_k (-1)^k \binom{n}{k} f(k) = (-1)^n \Delta^n f(0)$, the n-th finite difference. Useful fact: Δ^n annihilates degree-$< n$ polynomials, which is the boxed statement.

Partial alternating sums (these do have closed form)

$$\sum_{k=0}^{m} (-1)^k \binom{n}{k} = (-1)^m \binom{n-1}{m} \qquad \sum_{k=0}^{m} (-1)^k \binom{n+k}{k} = (-1)^m \binom{n+m+1}{m} \quad \text{(via upper negation + hockey stick)}$$

Useful specific alternating sums

$$\sum_k \frac{(-1)^k}{k+1}\binom{n}{k} = \frac{1}{n+1} \qquad \sum_k \frac{(-1)^k}{k+m}\binom{n}{k} = \frac{1}{m\binom{n+m}{m}} \qquad \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k}\binom{n}{k} = H_n = \sum_{i=1}^{n} \frac{1}{i}$$

$$\sum_k (-1)^k \binom{n}{k}\binom{k}{m} = (-1)^m [n=m]$$

$$\sum_k (-1)^k \binom{n}{k}(n-k)^n = n! \qquad \sum_k (-1)^k \binom{n}{k}(n-k)^m = 0 \text{ for } m < n$$

That last pair is the surjection count in disguise, and it's the standard way to compute Stirling numbers.

Binomial Inversion (Generalized PIE)

Not a forward simplification, but the primary way to *create* a solvable binomial sum when converting "at least k" to "exactly k":

$$f(n)=\sum_{k=0}^{n}\binom{n}{k}g(k)\iff g(n)=\sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}f(k)\qquad f(k)=\sum_{i=k}^{n}\binom{i}{k}g(i)\iff g(k)=\sum_{i=k}^{n}(-1)^{i-k}\binom{i}{k}f(i)$$

4. Polynomial Weights The Falling-Factorial Route

Never expand k^m directly. Convert to falling factorials via Stirling numbers of the second kind:

$$k^m=\sum_j S(m,j)\,k^{\underline{j}}$$

Then absorption kills each term. The master results:

$$\sum_k\binom{n}{k}k^{\underline{m}}=n^{\underline{m}}2^{n-m}\qquad\boxed{\sum_k\binom{n}{k}k^m=\sum_j S(m,j)\,n^{\underline{j}}2^{n-j}}$$

That's $O(m^2)$ total, independent of n so $\sum_k k^{100}\binom{n}{k}$ for $n=10^{18}$ is cheap.

Small cases to have memorized:

$$\sum_k\binom{n}{k}=2^n\qquad\sum_k k\binom{n}{k}=n2^{n-1}\qquad\sum_k k^2\binom{n}{k}=n(n+1)2^{n-2}\qquad\sum_k k^3\binom{n}{k}=n^2(n+3)2^{n-3}$$

With an extra x^k (differentiate $(1+x)^n$):

$$\sum_k\binom{n}{k}x^k=(1+x)^n\qquad\sum_k k\binom{n}{k}x^k=nx(1+x)^{n-1}\qquad\sum_k k^2\binom{n}{k}x^k=nx(1+x)^{n-2}(1+nx)$$

And the division direction (integrate or use downward absorption):

$$\sum_k\frac{1}{k+1}\binom{n}{k}=\frac{2^{n+1}-1}{n+1}\qquad\sum_k\frac{x^{k+1}}{k+1}\binom{n}{k}=\frac{(1+x)^{n+1}-1}{n+1}$$

5. Row Prefix Sums No Closed Form, But a Recurrence

$$S(n,k)=\sum_{i=0}^{k}\binom{n}{i}$$

has no closed form. Stop looking. But it satisfies:

$$S(n,k)=2S(n-1,k)-\binom{n-1}{k}\qquad S(n,k+1)=S(n,k)+\binom{n}{k+1}$$

so you can move $(n,k)\to(n\pm1,k\pm1)$ in $O(1)$ which means Mo's algorithm on queries answers q queries of $S(n_i,k_i)$ in $O((n+q)\sqrt{q})$. This is a known CP trick and it's the intended solution whenever you see "many queries of a partial binomial row sum."

6. Roots of Unity Filter Residue-Class Sums

To extract terms where $k\equiv r\pmod m$:

$$\sum_{k\equiv r\pmod m}\binom{n}{k}=\frac{1}{m}\sum_{j=0}^{m-1}\omega^{-jr}(1+\omega^j)^n,\qquad\omega=e^{2\pi i/m}$$

General form, for any $f(x)=\sum a_kx^k$:

$$\sum_{k\equiv r\pmod m}a_k=\frac{1}{m}\sum_{j=0}^{m-1}\omega^{-jr}f(\omega^j)$$

In CP mod p: you need an m-th root of unity mod p, which exists iff $m\mid p-1$. For $p=998244353$, $p-1=2^{23}\cdot7\cdot17$, so $m\in\{2,4,8,\dots,7,17,\dots\}$ all work. For $m=2$ it's trivial and needs no roots at all:

$$\sum_{k\text{ even}}\binom{n}{k}=\frac{2^n+0^n}{2}=2^{n-1}\ (n\geq1)\qquad\sum_{k\text{ odd}}\binom{n}{k}=2^{n-1}\qquad\sum_k(-1)^k\binom{n}{k}x^k=(1-x)^n$$

If $m\nmid p-1$, work in $\mathbb{F}_p[x]/(x^m-1)$ represent values as length-m vectors and multiply cyclically. That's the standard escape hatch.

7. Generating Function Dictionary

This is what you fall back on when nothing matches. Write your sum as a coefficient extraction, multiply the series, read off the coefficient.

Series	Coefficient of x^k	Series	Coefficient of x^k
$(1+x)^n$	$\binom{n}{k}$	$\frac{1-\sqrt{1-4x}}{2x}$	Catalan C_k
$\frac{1}{(1-x)^n}$	$\binom{n+k-1}{k}$	$\frac{1}{\sqrt{1-4x}}$	$\binom{2k}{k}$
$\frac{1}{1-x}$	1 (multiplying = prefix sums)	$\frac{1}{\sqrt{1-4x}}\left(\frac{1-\sqrt{1-4x}}{2x}\right)^a$	$\binom{2k+a}{k}$
$\frac{x^a}{(1-x)^{b+1}}$	$\binom{k-a+b}{b}$	$\frac{x}{1-x-x^2}$	Fibonacci F_k
$\frac{1-x^{c+1}}{1-x}$	1 for $0 \le k \le c$ (capped var)	$(e^x-1)^m/m!$ (EGF)	$S(k,m)$, Stirling 2nd kind
e^x (EGF)	$1/k!$	$\exp(e^x-1)$ (EGF)	Bell numbers $B_k/k!$

Bivariate Grid Path Extractors:

$$[x^n y^m]\frac{1}{1-x-y} = \binom{n+m}{n} \qquad [x^n y^m]\frac{1}{1-x-y-xy} = D(n,m) \text{ (Delannoy)}$$

Lagrange Inversion (when a tree/walk GF satisfies $A(x) = x\phi(A(x))$):

$$[x^n]A(x) = \frac{1}{n}[u^{n-1}]\phi(u)^n \qquad [x^n]A(x)^k = \frac{k}{n}[u^{n-k}]\phi(u)^n$$

The method ("snake oil"): given $\sum_k a_{n,k}$, define $F(x) = \sum_n (\text{your sum})x^n$, swap the order of summation so the inner sum is over n, recognize the inner generating function, simplify, and read the coefficient back off. This finds identities rather than looking them up, and it's the right tool when your sum isn't in any table.

Capped stars and bars via GF:

$$[x^K]\left(\frac{1-x^{M+1}}{1-x}\right)^N = \sum_j (-1)^j \binom{N}{j}\binom{K-j(M+1)+N-1}{N-1}$$

8. Convolution Identities Worth Recognizing

$$\sum_k \binom{2k}{k}\binom{2n-2k}{n-k} = 4^n \qquad \sum_k C_k C_{n-k} = C_{n+1} \quad \text{(Catalan)}$$

$$\sum_k \binom{n-k}{k} = F_{n+1} \qquad \sum_k \binom{n}{k} F_k = F_{2n} \qquad \sum_k \binom{n}{k}\binom{n}{k+m} = \binom{2n}{n+m}$$

$$\sum_k \binom{n}{k} 2^k = 3^n \qquad \sum_k \binom{n}{k} a^k b^{n-k} = (a+b)^n$$

$$\sum_k \binom{m}{k}\binom{n+k}{p} = \sum_j \binom{n}{p-j}\binom{m}{j}2^{m-j} \quad \longrightarrow \quad \text{expand } \binom{n+k}{p} \text{ by Vandermonde first, then swap sums}$$

That last line is the expansion direction, and it's the move people miss. Sometimes you must break one binomial into a convolution to create a Vandermonde shape. $\binom{a+b}{r} = \sum_k \binom{a}{k}\binom{b}{r-k}$ read left-to-right is how Codeforces 785D goes from $O(n^2)$ to $O(n)$.

9. Named Families (Recognize Them From the Shape)

Object	Formula	Signature
Catalan C_n	$\frac{1}{n+1}\binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1}$	balanced brackets, non-crossing, stay ≥ 0
Ballot / Gen. Catalan	$\frac{a-b}{a+b}\binom{a+b}{a}$	strictly-ahead paths
m-ary trees	$\frac{1}{mn+1}\binom{mn+1}{n}$	the $\frac{1}{n}$ fingerprint $\to$ cycle lemma / Lagrange
Narayana	$\frac{1}{n}\binom{n}{k}\binom{n}{k-1}$	Catalan refined by number of peaks
Motzkin M_n	$M_n = \sum_k \binom{n}{2k} C_k$	paths with a flat step allowed
Delannoy $D(m,n)$	$D(m,n) = \sum_k \binom{m}{k}\binom{n}{k} 2^k$	grid paths with diagonal steps
Schröder	$\sum_k \binom{n}{k}\binom{n+k}{k}\frac{1}{k+1}$	diagonal lattice paths below a line
Derangements D_n	$D_n = n!\sum_{k=0}^{n}\frac{(-1)^k}{k!}$, $D_n = (n-1)(D_{n-1}+D_{n-2})$	permutations with no fixed points
Surjections onto $[k]$	$k!\,S(n,k) = \sum_j (-1)^j \binom{k}{j}(k-j)^n$	every box nonempty
Stirling 2nd $S(n,k)$	$S(n,k) = kS(n-1,k) + S(n-1,k-1)$	set partitions into k blocks
Stirling 1st $c(n,k)$	$c(n,k) = (n-1)c(n-1,k) + c(n-1,k-1)$	permutations with k cycles
Bell B_n	$B_{n+1} = \sum_k \binom{n}{k} B_k$	all set partitions
Eulerian $A(n,k)$	$A(n,k) = (k+1)A(n-1,k) + (n-k)A(n-1,k-1)$	permutations with k descents
Partitions $p(n)$	pentagonal recurrence, $O(n\sqrt{n})$	unordered sums

Two conversion identities that connect powers and falling factorials (how you move between "polynomial in k" worlds), plus **Worpitzky's Identity**:

$$x^n = \sum_k S(n,k)x^{\underline{k}} \qquad x^{\underline{n}} = \sum_k (-1)^{n-k} c(n,k) x^k \qquad x^n = \sum_{k=0}^{n} A(n,k)\binom{x+k}{n}$$

Worth knowing: Worpitzky's identity turns $\sum_{x=0}^{N} x^n$ into a hockey-stick sum in $O(n^2)$ and handles $\sum_{i=1}^{n} i^m x^i$ for huge n.

10. The NT $\times$ Combinatorics Crossovers

$$\sum_{d|n}\varphi(d) = n \qquad \#\{i \leq n : \gcd(i,n) = d\} = \varphi(n/d)$$

$$\text{Necklaces: } \frac{1}{n}\sum_{d|n}\varphi(n/d)k^d \qquad \text{with exactly } m \text{ ones: } \frac{1}{n}\sum_{d|\gcd(n,m)}\varphi(d)\binom{n/d}{m/d}$$

$$\sum_{i_1,\ldots,i_r \leq M}[\gcd(i_1,\ldots,i_r) = 1] = \sum_{d=1}^{M}\mu(d)\lfloor M/d\rfloor^r$$

$$v_p\left(\binom{n}{r}\right) = \#\text{carries adding } r+(n-r) \text{ in base } p \quad \text{(Kummer)}$$

$$\binom{n}{r} \bmod p = \prod_i \binom{n_i}{r_i} \bmod p \quad \text{(Lucas)} \qquad \binom{n}{r} \text{ odd} \iff (r \;\&\; n) = r$$

11. Rarer But Occasionally Decisive

$$\sum_k \binom{n}{k} x(x+k)^{k-1}(y+n-k)^{n-k} = (x+y+n)^n \qquad \text{(Abel's Identity)}$$

$$\sum_k (-1)^k \binom{2n}{n+k}^3 = \frac{(3n)!}{(n!)^3} \qquad \text{(Dixon's Identity)}$$

$$\sum_k \binom{n}{k}^3 \longrightarrow \text{No closed form, } O(n) \text{ via P-recursive recurrence} \qquad \text{(Franel Numbers)}$$

$$\sum_k \binom{n}{k}\binom{n+k}{k}(-1)^k = (-1)^n \qquad \text{(Delannoy Diagonal)}$$

If a sum has no closed form, it is often still **holonomic / P-recursive**: the terms satisfy a linear recurrence with polynomial coefficients, computable in $O(n)$ (e.g., Franel numbers satisfy $(n+1)^2F_{n+1} = (7n^2+7n+2)F_n + 8n^2F_{n-1}$). Zeilberger's algorithm finds it; in contest you'd guess the recurrence from brute-forced terms (via Gaussian elimination or Berlekamp-Massey variants) and verify. That's a legitimate and underused CP tactic.

How to Actually Use This

1. **Write the brute force as nested sums.** Don't try to be clever yet.
2. **Swap the order of summation.** The inner sum should be over the index that appears in the most places.
3. **Kill loose polynomial factors with absorption** (convert k^m to falling factorials first; use downward absorption for $\frac{1}{k+1}$).
4. **Look at the indices.** Lower indices sum to a constant $\to$ Vandermonde (§1). One counts up while the other counts down (or upper indices sum to a constant) $\to$ split-a-line (§2). A single index counting up a column $\to$ hockey stick (§2).
5. **If it's alternating, check the polynomial degree first** it may just be zero (§3).
6. **If nothing matches, go to generating functions (§7)** and read off a coefficient.
7. **If still nothing, brute force** $n \leq 8$ and search OEIS, or guess a P-recursive recurrence from the terms.

Verification Discipline: Every closed form you derive gets checked against brute force on all small inputs before you submit. Boundary cases in these identities (empty sums, $k > n$, $n = 0$) are where they silently break, and a 60-second comparison loop catches every one of them.